Departamento de Economía
Universidad de Concepción

INCAR²
Interdisciplinary Center for Aquaculture Research - Applied Research

The Impact of Environmental Variability on Fishers' Harvest Decisions in Chile

A multi-species structural approach

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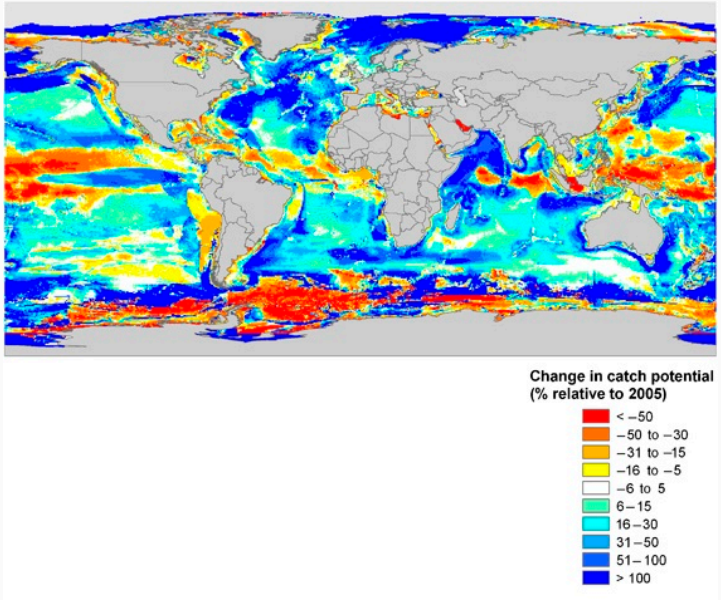
ICES/PICES Small Pelagic Symposium — La Paz, México, May 8, 2026

Funded by ANID-Chile, FONDECYT Iniciación N° 11250223.

Motivation

- Small pelagic stocks fluctuate sharply with climate variability.
- **Standard projection tools assume constant fishing rates or full quota uptake** — the human decision is assumed away.
- But fishers **choose**: which species to target, how many trips, how much to land.



Source: Cheung et al. (2010)

> ***Project goal:*** *propagate climate-driven biomass shifts into fisher-side responses — participation today (Paper 1),* ***optimal harvest decisions*** *in the full structural model (Paper 2).*

The project — two papers, one structural pipeline

Paper 1 (today's results) *Reduced-form effort + identified stock dynamics*

- **Negative binomial** trip equation, estimated separately by fleet.
- Bayesian state-space stock dynamics with structural climate shifters (in appendix).
- Comparative statics under future climate scenarios.

Paper 2 (today's roadmap) *Welfare counterfactuals*

- Translog **cost** function (trip-level, by fleet).
- **Inverse demand** (IAIDS, 3SLS).
- **Dynamic vessel optimization** $\Rightarrow$ optimal $T_g, h_{g\tau}$ under different climate scenarios.
- Welfare, prices, **species substitution**.

> *The two papers share the **same identified stock dynamics**. Paper 1 stops at participation; Paper 2 closes the loop through prices, cost and profits to estimate optimal harvest and quota allocation.*

Setting and observed heterogeneity

Chile's Small Pelagic Fishery (Centro-Sur)

Stocks & sectors

- **Anchoveta** (*Engraulis ringens*), **sardine** (*Strangomera bentincki*), **jack mackerel** (*Trachurus murphyi*).
- ~94% of national landings; mostly purse seiners.
- Two sectors: **artisanal** (ART), **industrial** (IND).

Quota regime

- TAC split between sectors with **limited cross-sector transferability**.
- IND quotas partially tradable within sector; ART under spatial collective-property rights (i.e. **RAE**).

Fleet portfolios are very different

- **ART**: ~ {anchoveta, sardine}. Concentrated.
- **IND**: ~ 95% jack mackerel + 5% sardine.

Mobility

- IND: offshore purse-seiners.
- ART: port-anchored, RAE binds spatially.

$\Rightarrow$ Same shock, **different choices**.

Observed strategy switching — ART vs IND

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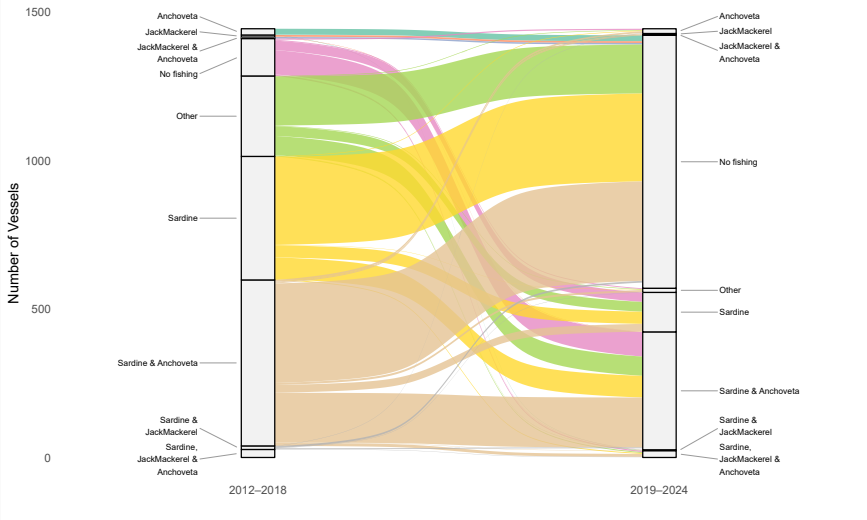
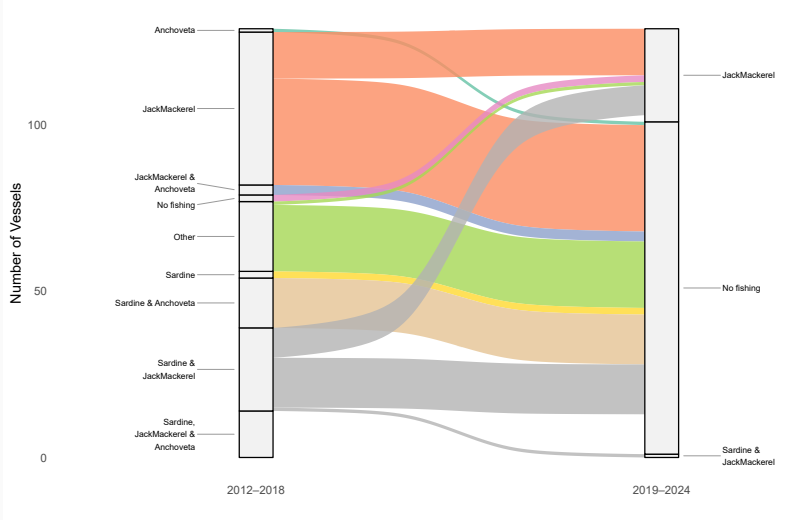
Artisanal



Industrial



Trip-level target-species transitions, IFOP logbooks 2013–2024.

> *ART: anchoveta↔sardine dominates; jack mackerel essentially absent. IND: rotation across all three. Same fishery, very different revealed choice sets.*

Paper 1 — The NB trip equation

A negative binomial model of annual trips

Specification, by fleet g (ART, IND)

$$T_{vy}^{g} \sim \mathrm{NB}\big(\exp(U_{vy}'\beta_g + \alpha_y^g),\ \theta_g\big)$$

with $U_{vy} = \big[\, p_{sy},\ H_{vy}^{alloc},\ Z_v,\ O_{vy} \,\big]$:

- p_{sy}: ex-vessel prices, by species.
- H_{vy}^{alloc}: vessel-level quota allocation.
- Z_v: hold capacity, vessel type.
- O_{vy}: bad-weather days, closures.
- α_y^g: **year FE** (absorbs 2019–22 shocks).
- Vessel-clustered robust SEs.

Why NB by fleet?

- Overdispersion σ^2/μ: **22.4** (ART), **4.9** (IND). Poisson rejected at $p < 0.001$.
- Two fleets, two technologies, two choice sets — no pooling.

Two channels to projections

- **Indirect**: structural shifters $\rho_i^{SST}, \rho_i^{CHL} \to B_{i,t} \to H_{vy}^{alloc}$. *Jurel: testing ρ_{ENSO} over Niño 3.4 instead* (App. D-bis).
- **Direct**: vessel-specific Δbad-weather days from CMIP6 winds.

What drives fisher participation?

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Variable	Industrial	Artisanal	Reads as
H_{vy}^{alloc}	**(+)** ***	**(+)** ***	Quota = binding budget constraint, both fleets.
Price — jack mackerel	**(+)** ***	0	IND oriented to jack mackerel; ART has no exposure.
Price — sardine	0	**(+)** ***	Sardine is ART's primary target.
Price — anchoveta	0	**(–)** ***	Simultaneity sign.
Hold capacity (log)	0	**(+)** ***	Heterogeneity is *within* the artisanal fleet.
Bad-weather days	0	**(–)** ***	ART vulnerable; IND insulated by vessel size.
Closure days	**(–)** ***	**(+)** ***†	IND: largest single coef. ART: zone FE collinear.

Negative binomial **with year FE**, vessel-clustered robust SEs. Sign and significance shown. * $p < 0.01$. † zone-FE collinear (not causally interpretable).

> *Two fleets, **completely different participation models**.*

Comparative statics — the fleet asymmetry

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Fleet	Counterfactual	$\%\Delta T$ marginal	$\%\Delta T$ conditional	$\Pr(\text{portfolio loss} > 50\%)$
Artisanal	mid-century, moderate (SSP2-4.5)	**–8.1%**	–2.1%	**0.95**
	end-century, severe (SSP5-8.5)	**–10.2%**	–2.7%	**0.99**
Industrial	mid-century, moderate (SSP2-4.5)	–0.7%	–0.7%	0.12
	end-century, severe (SSP5-8.5)	–0.9%	–0.8%	0.12

$\%\Delta T$: change in annual trips vs baseline. **Marginal** averages over all posterior draws. **Conditional** restricts to draws with $f_v^H > 0.5$. Cross-median across a 6-model CMIP6 ensemble. NB with year FE.

- Marginal asymmetry ~**11:1** ART:IND; conditional ~**3.4:1**.
- ART portfolio collapse near-certain by SSP5-8.5 end.
- $\Rightarrow$ Asymmetry runs through **probability of portfolio collapse** $\times$ **direct weather** ($\beta_w^{ART} < 0$, $\beta_w^{IND} = 0$); quota cross-sector limits lock it in.
- IND's protection rests on jack mackerel (long-run *local* shifter **n.i.**; ENSO test in progress, App. D-bis).

Two channels: direct vs indirect

Decomposition

Indirect (biomass → allocation):

$$\Delta SST,\ \Delta \log CHL \xrightarrow{\rho_i^{SST},\ \rho_i^{CHL}} B_{i,t}^{\star} \rightarrow H_{vy}^{alloc}$$

Jurel: extension to ρ_{ENSO} over Niño 3.4 in progress (App. D-bis).

Direct (weather → trips):

$$\Delta\text{wind} \xrightarrow{\text{vessel CDF}} \Delta\text{bw-days}_{vy} \xrightarrow{\beta_w^g} T_{vy}$$

End-century SSP5-8.5

Fleet	Indirect	Direct	Total
Artisanal	−9.0 pp	−1.2 pp	**−10.2%**
Industrial	−0.8 pp	−0.1 pp	**−0.9%**

> *Indirect (biomass) carries **most** of the asymmetry; direct (weather) reinforces it where vessels are exposed (ART). Both channels point the same way.*

Paper 2 — closing the loop

Three structural extensions ⇒ welfare

What gets added on top of Paper 1

(i) Trip-level cost function — translog $C_{vg}(w, h, x, Z, Env)$, SUR by fleet. Recovers **input demands** (fuel, labor, ice) and elasticities of input substitution under climate shocks.

(ii) Inverse demand (IAIDS) — $\ln p_{iy}$ as a function of all species' landings, instrumented by SST, CHL, fuel prices. Resolves the simultaneity observed in paper 1.

(iii) Vessel dynamic optimization — $\max_{T_g, h_{g\tau}} \sum_\tau \delta^\tau [P(h)\, h - C_g(h)]$ s.t. quota cap. $(T_g^*, h_{g\tau}^*)$ under each scenario.

Optimization model allows for

(i) Prediction under a changing climate of: Equilibrium prices and species substitution

(ii) Define optimal quota allocation

(iii) Study policy reforms (i.e., contrafactuals) as adaptation:

- Cross-sector quota transferability.
- RAE redesign (multi-zone permits).
- State-contingent TACs.

Takeaways

- **Paper 1**: fleet-specific NB model of annual trips reveals **two completely different participation technologies** in the same fishery.
- **Paper 1**: under counterfactual shocks, the marginal ART:IND participation asymmetry is **~11:1** (conditional **~3.4:1**), running through (i) the **probability of portfolio collapse** in the indirect biomass channel and (ii) **direct weather** exposure ($\beta_w^{ART} < 0$, $\beta_w^{IND} \approx 0$); quota cross-sector limits lock it in.
- **Paper 2**: closes the loop — **input demand, inverse demand, vessel × social-planner joint optimization** — to deliver welfare counterfactuals: equilibrium prices, species substitution & optimal quota allocation, and policy reforms.

> *Adaptive capacity is not a property of fishers; it is a property of the* ***bundle (portfolio × permit × institution × market)*** *they sit in. Identifying adaptation requires modeling all four dimensions jointly — pooling, fixed prices, or aggregate effort all bias the answer.*

¡Gracias! / Thank you

Questions?

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Appendix

Appendix A · Stock dynamics — spec & priors

Process (Schaefer with shifter on r):

$$B_{i,t+1} = B_{i,t} + r_{i,t} B_{i,t}\left(1 - \frac{B_{i,t}}{K_i}\right) - C_{i,t} + \varepsilon_{i,t}$$

with $r_{i,t} = r_i^0 \exp(\rho_i^{SST} \Delta SST_{t-1} + \rho_i^{CHL} \Delta \log CHL_{t-1})$.

Stock-specific extension in progress: jurel forced by ρ_{ENSO} over Niño 3.4 instead of local $(SST_{D1}, \log CHL_{D1})$ — see App. D-bis.

Observation (log-normal): $\log B_{i,t}^{obs} = \log B_{i,t} + u_{i,t}$.

Three nested specs: **ind**, **omega**, **full**.

Priors:

- $r_i^0, K_i, \sigma_{proc}, \sigma_{obs}$ from IFOP / SPRFMO single-species assessments.
- $\rho_i^{SST}, \rho_i^{CHL} \sim \mathcal{N}(\hat{\rho}_i^{stress}, 1)$.
- Ω: LKJ.

Estimation: Stan / HMC, 4 chains $\times$ 2000 post-warmup; $\hat{R} < 1.01$, $ESS > 400$.

Appendix B · Climate shifters

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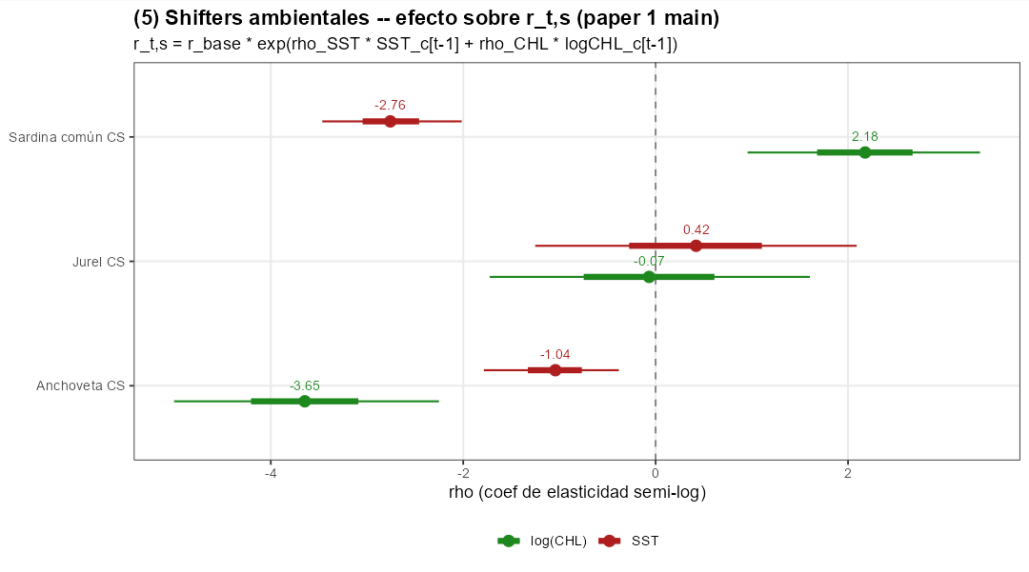

Posterior densities of ρ^{SST} and ρ^{CHL}, full multi-species specification. Anchoveta and common sardine: priors dominated by data. Jack mackerel: posteriors track priors $\rightarrow$ **n.i.** at the *local* coastal scale — **basin-scale ENSO shifter being tested next** (App. D-bis).

Appendix C · Stock productivity comparative statics

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Stock	Scenario	$\Delta r/r_0$ (cross-median)	90% CI	$\Pr(\Delta < 0)$
Anchoveta	SSP2-4.5, mid-century	**–51%**	[–68%, –27%]	1.00
	SSP5-8.5, end-century	**–90%**	[–98%, –54%]	0.99
Sardine	SSP2-4.5, mid-century	**–79%**	[–86%, –67%]	1.00
	SSP5-8.5, end-century	**–100%**	[–100%, –99%]	1.00
Jack mackerel	all scenarios	n.i.	n.i.	n.i.

Cross-median across CMIP6 ensemble of 6 GCMs (CESM2, CNRM-ESM2-1, GFDL-ESM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, UKESM1-0-LL); 90% CI is the within-posterior credible interval averaged across models. *Jurel: local* $(SST_{D1}, \log CHL_{D1})$ *shifter n.i. — basin-scale ENSO refit in progress (App. D-bis).*

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non-identification $\neq$ no climate effect

Two readings to keep separate:

- **Non-identification at the CS scale** — what the structural fit reports. With $N = 16$ informative obs and $\hat{\sigma}_{SST,D1} \approx 0.26°\mathrm{C}$, only large *local* elasticities would be detectable.
- **Climate insensitivity of jack mackerel** — *not* what the result implies. Peña-Torres et al. (2017) (ENSO & location choices) and Arcos et al. (2001) (SPRFMO-scale dynamics) document that climate *does* affect this stock at scales beyond local Chilean coastal anomalies.

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non-identification $\neq$ no climate effect

Three convergent tests of the *local* null (CS scale, coastal forcing):

(i) Spatial-domain robustness. Re-fits over 3 nested coastal domains: $\sigma_{post}/\sigma_{prior}$ for ρ^{SST}_{jurel} stays in [0.998, 1.014]. Anch & sard stable (CV < 0.03).

(ii) Dual-source biomass. Augment with IFOP Northern Chile acoustic series (2010–24, $r_{log} = 0.88$): $\sigma_{post}/\sigma_{prior} \in \{0.94, 0.99\}$. Bottleneck is **structural**, not N.

(iii) OROP-PS coherence. SPRFMO transzonal index correlates with CS at $r = 0.11$ over $N = 17$ — range-wide forcing cannot pin it down either.

$\Rightarrow$ The *local* null is robust. Natural reading: **relevant climatic drivers operate at scales beyond CS coastal forcing** — consistent with a transboundary species. Tested next on the obvious candidate.

Appendix D-bis — ENSO as basin-scale shifter (spec)

Hypothesis

Jurel CS responds to **ENSO via teleconnections**, not local CS coastal anomalies.

- Anch / sard: unchanged.
- **Jurel:** replace $(SST_{D1}, \log CHL_{D1})$ by single shifter ρ_{ENSO} over Niño 3.4.
- Lags $t-1$ (main), $t-2$ (sensitivity).

Pipeline

- Historical Niño 3.4: OISST / ERSST.
- CMIP6: 6-GCM ensemble re-extracted over Niño 3.4.
- Same Stan architecture as App. E (stock-specific shifters).
- Prior: $\rho_{ENSO} \sim \mathcal{N}(0, 0.5)$ — $1\sigma_{ENSO}$ shock $\Rightarrow \leq 65\%$ change in r.

> *Status: data extraction in progress; Stan refit by mid-May.*

Appendix D-ter — ENSO: asymmetric bet on the result

Scenario A — identifies

$\sigma_{post}/\sigma_{prior} \leq 0.7$:

- Null becomes **basin-scale identification**.
- Asymmetry **11:1 recomputed** with structural base for *all 3* stocks.
- Direct test of Peña-Torres 2017 / Arcos 2001 mechanism.

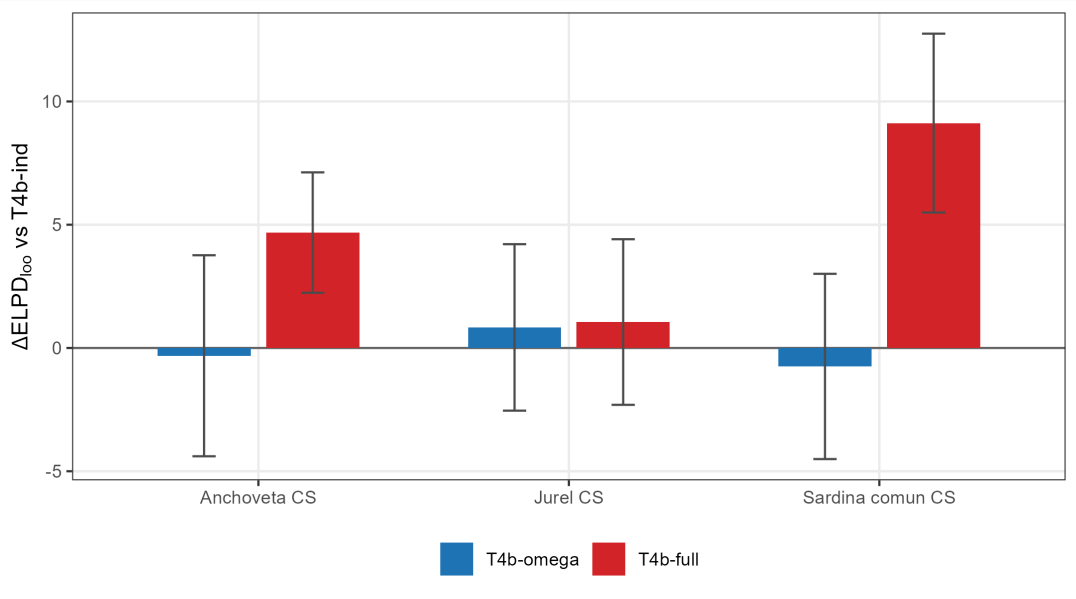
Scenario B — not identified

$\sigma_{post}/\sigma_{prior} \approx 1$:

- Null moves from 3 to **4 converging lines of evidence**.
- Climate effects on jurel operate on *margins* (location, behavior), not local productivity.

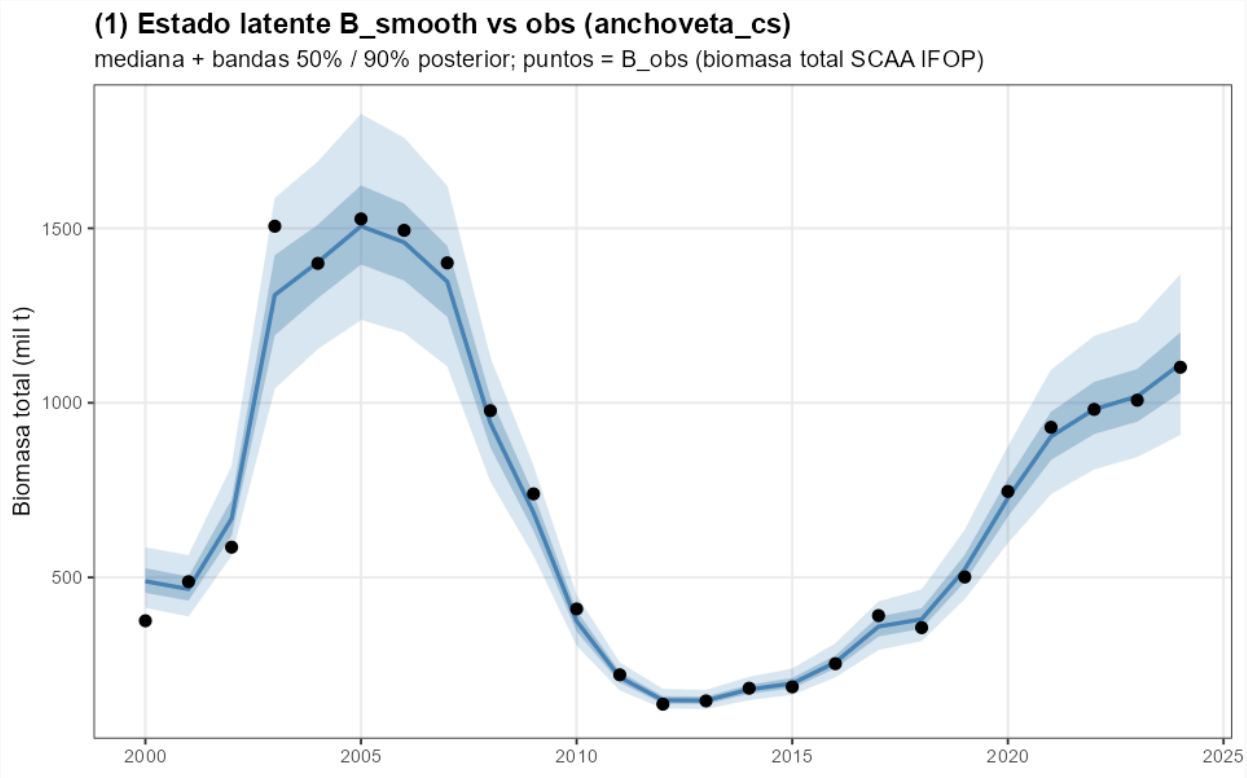
> → *Either scenario improves the paper.*

Appendix E · Model comparison (LOO)



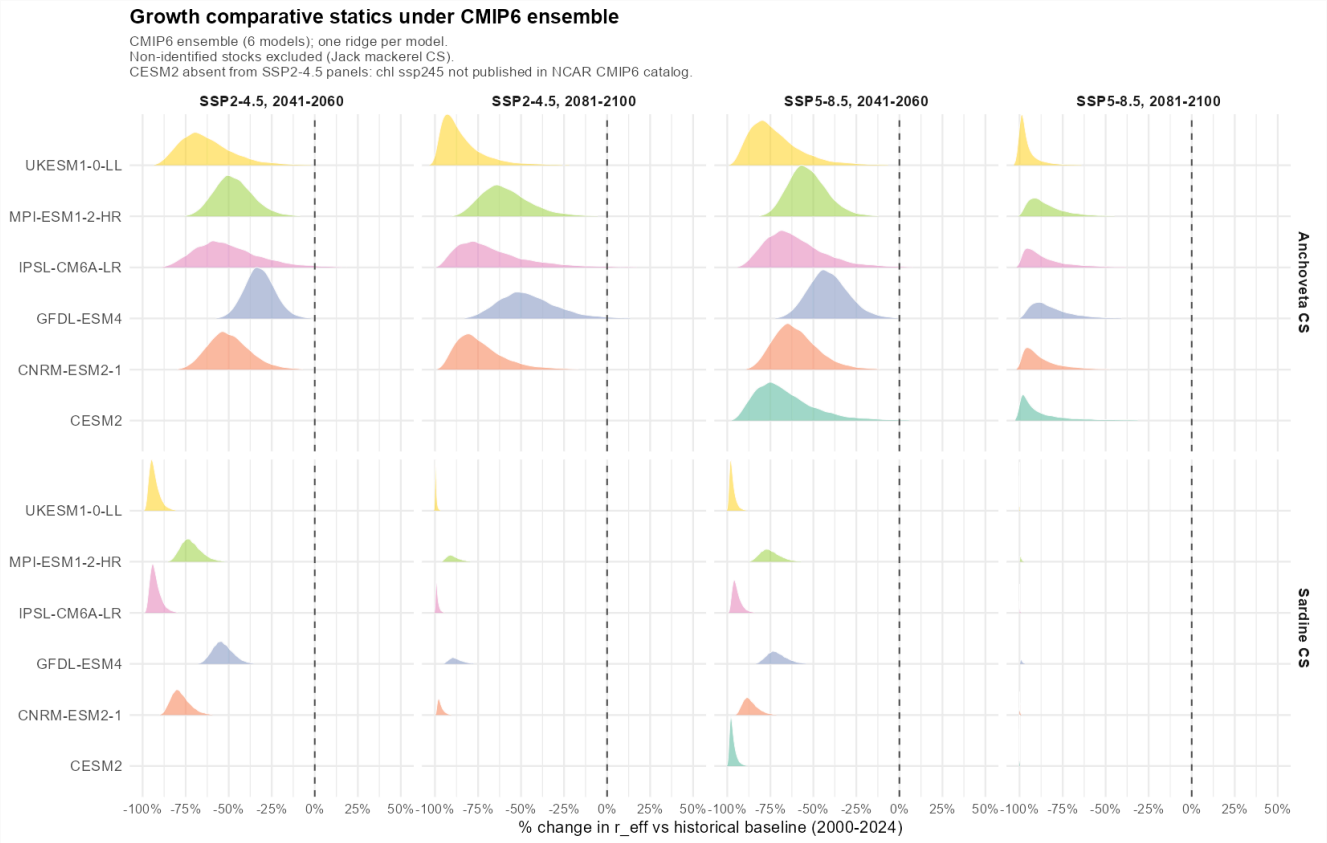
LOO ELPD across specs: **ind** (Ω diagonal, $\rho = 0$), **omega** (Ω full, $\rho = 0$), **full** (Ω full, $\rho \neq 0$). Full wins at $\Delta\text{ELPD} \approx 14$–$18$; LFO one-step-ahead essentially tied.

Appendix F · Posterior predictive check



Smoothed posterior of biomass vs observed acoustic estimates, full model.

Appendix G · CMIP6 ensemble — ridgeline



Posterior $\Delta r/r_0$ across SSP $\times$ period combinations, one ridge per CMIP6 model (6-GCM ensemble).

Appendix H · Variance decomposition of trip response

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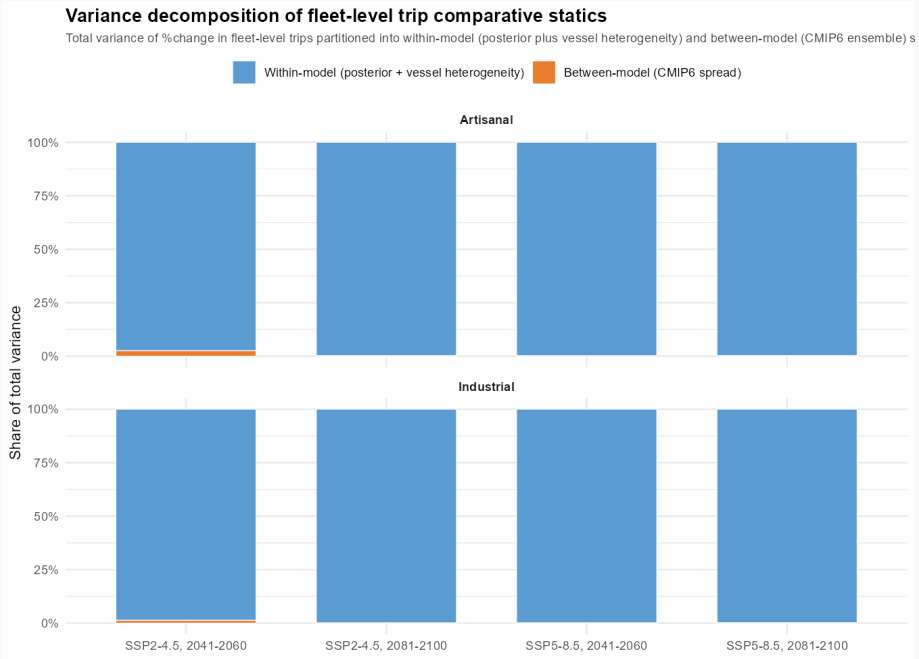

- **Within-model** (posterior + vessel heterogeneity) dominates: **97%–100%** across all SSP $\times$ period $\times$ fleet cells (re-computed after the direct weather channel was added).
- **Between-model** (CMIP6 ensemble): at most **3%**, often ≈ 0.
- The narrow cross-IQR in Table 5 is *not* climate-model consensus — it is a **floor effect**: once $f_v^H \to 0$, $\%\Delta T \to \exp(-\beta_g H_v^{alloc})$ — a fleet-mechanical quantity, not a climate magnitude.

Appendix I · Paper 2 — data inputs

- Trip-level catch and price data (IFOP logbooks, 2013–2024).
- All-gear official catch series (SERNAPESCA).
- Vessel characteristics & ports of operation.
- Environmental covariates (Copernicus, GlobColour, CMIP6 6-GCM ensemble).

> ***Caveat — IFOP panel coverage*** *covers the* ***purse-seine artisanal*** *subset; non-PS gears (lampara, lines) excluded.*